

Dioscorea villosa (Dioscoreaceae)

Wild Yam, Colic Root

Wild yam is the plant source of a steroid-like substance, diosgenin, which was the starting point in the creation of the first contraceptive pill. There is no suggestion that the plant was used as a contraceptive in the past, though it has traditionally been taken in Central America to relieve menstrual, ovarian, and labor pains. The herb is also valuable for digestive problems, arthritis, and muscle cramps.



Wild yam can be found growing wild in damp woodlands in North America.

Habitat & Cultivation

Wild yam is native to North and Central America, and has now become naturalized in tropical, semitropical, and temperate climates around the world. The plant is propagated from seed in spring, or from sections of tubers or by root division in spring or autumn. It thrives in sunny conditions and rich soil. The root and tuber of wild yam are harvested in autumn.

Related Species

Many yam species have a hormonal action. *Shan yao* (*Dioscorea opposita*) is an important tonic for the stomach and digestion in traditional Chinese medicine, and is taken for appetite loss and wheezing.

Key Constituents

- Steroidal saponins (mainly dioscin)
- Phytosterols (beta-sitosterol)
- Alkaloids
- Tannins
- Starch

Key Actions

- Antispasmodic
- Anti-inflammatory
- Antirheumatic
- Increases sweating
- Diuretic

Research

■ Synthesis of hormones

Diosgenin, a breakdown product of dioscin, was first identified by Japanese scientists in 1936. This discovery paved the way for the synthesis of progesterone (one of the main female sex hormones) and of corticosteroid hormones such as cortisone. Wild yam does not contain estrogen or progesterone, and wild yam "natural progesterone" products are derived from chemically processed diosgenin, and have little or no relationship with naturally occurring wild yam.

■ Anti-cholesterol activity

Research indicates that diosgenin reduces the absorption of cholesterol from the gut and increases its elimination from the body.

Traditional & Current Uses

■ **Traditional uses** Both the Maya and the Aztec peoples used wild yam medicinally—possibly to relieve pain. The plant is also known as colic root and rheumatism root in North America, indicating its use by European settlers for these conditions.

■ Gynecological problems

A traditional remedy for painful periods and ovarian pain, wild yam has estrogen-modulating activity and is used to treat menopausal symptoms.

■ Arthritis & rheumatism

The herb's combination of anti-

Parts Used

Root and tuber have valuable antispasmodic properties. They are used to treat colic and menstrual cramps.



Fresh root and tuber



Chopped dried root and tuber



Rhizome has muscle-relaxant properties

Dried root and tuber

Key Preparations & Their Uses

Caution Do not take during pregnancy.



Decoction (to make, p. 291).
For irritable bowel syndrome, take 1/3 cup (75 ml) twice a day.



Tincture (to make, p. 292).
For arthritis, take 1/2 tsp with water twice a day.

■ Digestive problems

The herb can be used as an effective treatment for digestive problems, including gallbladder inflammation, irritable bowel syndrome, and diverticulitis.

Self-help Use

- Period pain, p. 315.